Experiment-9

Write a python program to implement Travelling Salesman Problem

Aim

To write and execute a Python program to solve the Travelling Salesman Problem by finding the shortest route that visits every city exactly once and returns to the starting city.

Algorithm

1. Represent the distances between the cities using a distance matrix.
2. Start the tour from city A.
3. Generate all possible orders (permutations) of the remaining cities.
4. For each route, calculate the total distance including the return to city A.
5. Compare the cost of the route with the best cost found so far.
6. If the cost is lower, store the route as the best route.
7. Repeat the process for all possible routes.
8. Display the shortest route and its minimum cost.

Coding

ex-9.py - D:/ex-9.py (3.10.0)

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```
# Travelling Salesman Problem (Brute Force)
from itertools import permutations

cities = ["A", "B", "C", "D"]

# Distance between every pair of cities
dist = [[0, 10, 15, 20],
        [10, 0, 35, 25],
        [15, 35, 0, 30],
        [20, 25, 30, 0]]

n = len(cities)
best_cost = 999999
best_route = None

print("TRAVELLING SALESMAN PROBLEM")
print("---------------------------")

# Start from city A and try every order of the other cities
for route in permutations(range(1, n)):
    cost = 0
    current = 0
    for city in route:
        cost = cost + dist[current][city]
        current = city
    cost = cost + dist[current][0]      # return to the start

    path = "A"
    for city in route:
        path = path + " -> " + cities[city]
    path = path + " -> A"
    print(path, " Cost =", cost)
```

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```
    if cost < best_cost:
        best_cost = cost
        best_route = path

print()
print("Shortest Route :", best_route)
print("Minimum Cost   :", best_cost)
```

Output

IDLE Shell 3.10.0

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```
Python 3.10.0 (tags/v3.10.0:b494f59, Oct  4 2021, 19:00:18) [MSC v.1929 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========================== RESTART: D:/ex-9.py ==========================
TRAVELLING SALESMAN PROBLEM
---------------------------
A -> B -> C -> D -> A  Cost = 95
A -> B -> D -> C -> A  Cost = 80
A -> C -> B -> D -> A  Cost = 95
A -> C -> D -> B -> A  Cost = 80
A -> D -> B -> C -> A  Cost = 95
A -> D -> C -> B -> A  Cost = 95

Shortest Route : A -> B -> D -> C -> A
Minimum Cost   : 80
>>>
```

Result

The Travelling Salesman Problem was successfully solved using the brute force method. The shortest route found is:

A -> B -> D -> C -> A

with a minimum cost of 80.